Linear programming problems:

Linear programming is a tool to study a broader subject called 'Operational Research'.

Operational research is a "scientific approach to decision making, which seeks to determine how best to design & operate a system, under conditions requiring the allocation of scarce resources".

Subject orgin was during W.W II, where scientists were asked to analyze military problems.

Now it is extensively studied in engineering, business, manufacturing.

Example: A furniture dealer deals with only 2 items - tables & chairs.

Her investment is ₹ 50,000, and strorage space is for 60 items.

A table cost ₹ 2500 & she can make profit of ₹ 250.

A chair cost ₹ 500 & she can make profit of ₹ 75.

Question: How many chairs & tables she should buy from the available money so as to maximise her total profit, assuming she can sell all items she buys?

Sol: x be no of tables that dealer buys

y be no of chairs —"—.

Obviously $x \geq 0$ & $y \geq 0$. —— (1)

Since maximum she can buy is 60 items,

$x + y \leq 60.$ —— 2

Also maximum amount she can invest is

₹ 50000 $\therefore$ $2500x + 500y \leq 50{,}000$ — 3

Dealer wants to invest in such a way so as to maximize her profit.

If Z is total profit, then, $Z = 250x + 75y$.

Question: maximize

$$Z = 250x + 75y$$

With constraints:

$$5x + y \leq 100$$

$$x + y \leq 60$$

$$x \geq 0, \; y \geq 0.$$

1. Graphical Method to solve LPP
(Best of corner method)

Question:

Maximize $Z = 3x_1 + 5x_2$ subjected to

$$x_1 + 2x_2 \leq 2000$$
$$x_1 + x_2 \leq 1500$$
$$x_1 \geq 0 \quad ; \quad x_2 \geq 0$$
$$x_2 \leq 600$$

<u>Solution</u>:

We draw lines for each inequalities assuming they are equal

ie $x_1 + 2x_2 = 2000$

Let $x_2 = 0$ then $x_1 = 2000$

$x_1 = 0$ then $x_2 = 1000$

$\therefore$ Points are $(2000, 0)$, $(0, 1000)$ $\longrightarrow l_1$

For $x_1 + x_2 = 1500$

$(0, 1500)$ & $(1500, 0)$ —— l_2

$x_1 = 0$ x_2 can be any no $- l_3$

$x_2 = 0$ x_1 can be any no $- l_4$

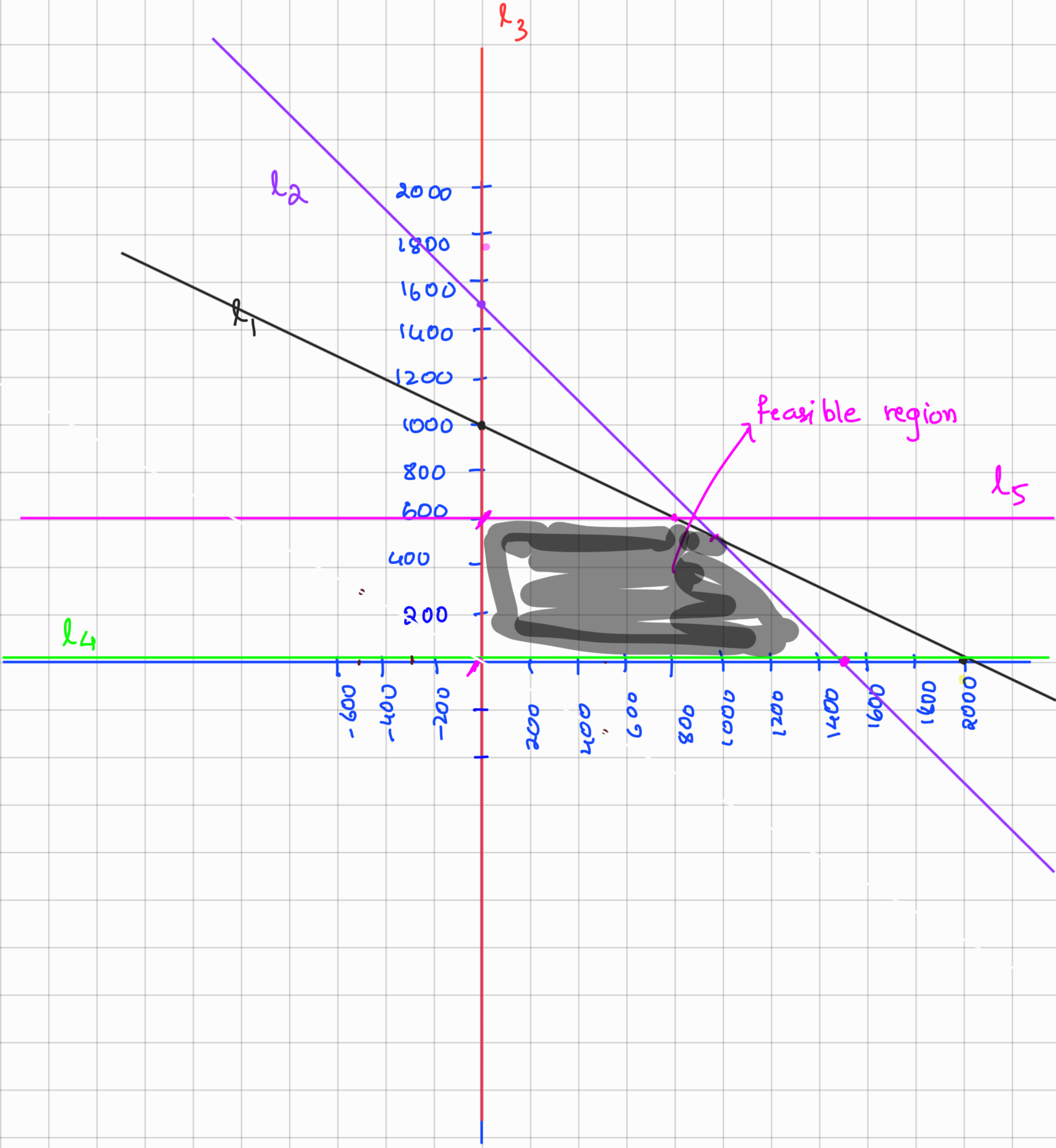
$x_2 = 600$ x_1 can be any no $\rightarrow l_5$

The intersecting points in the feasible region are (0, 600), (0, 0), (1500, 0), (800, 600), (1000, 500).

Substitute in $Z = 3x_1 + 5x_2$

	Z
(0, 600)	3000
(0, 0)	0
(1500, 0)	4500
(800, 600)	5400
(1000, 500)	5500

$\therefore$ Optimal point is (1000, 500) with maximum value $Z = 5,500$



l_3
l_2
l_1
l_4
l_5
feasible region
2000
1800
1600
1400
1200
1000
800
600
400
200
-600
-400
-200
200
400
600
800
1000
1200
1400
1600
1800
2000

Simplex method

Slack variable

A non-negative variable which is added to the left hand side of the constraint to convert $\leq$ to an equation is called a slack variable.

Example

$$3x_1 + x_2 \leq 2$$

$$x_1 + x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

Ans:

$$3x_1 + x_2 + s_1 = 2$$

$$x_1 + x_2 + s_2 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Surplus variable

A non-negative variable which is subtracted from the left hand side of the constraint to convert $\geq$ to an equation is called a surplus variable.

Example

$$3x_1 + x_2 \geq 2$$

$$x_1 + x_2 \geq 5$$

$$x_1, x_2 \geq 0$$

Ans:

$$3x_1 + x_2 - s_1 = 2$$

$$x_1 + x_2 - s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Standard form of LPP for applying simplex method

1. All the constraints should be converted to equations except for the non-negativity constraint which remains as inequalities.

Example

$$3x_1 + x_2 \leq 2$$

$$x_1 + x_2 \geq 5$$

$$x_1, x_2 \geq 0$$

Ans:

$$3x_1 + x_2 + s_1 = 2$$

$$x_1 + x_2 - s_2 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

2. Right side element of each constraint should be made non-negative, if not, it can be made positive on multiplying both side of the resulting equation by (-1).

Example

$$x_1 + x_2 \geq -5 \quad \longrightarrow \quad -x_1 - x_2 \leq 5$$

$$x_1, x_2 \geq 0 \qquad\qquad -x_1 - x_2 + s_1 = 5$$

Ans:

$$x_1 + x_2 - s_2 = -5$$

$$\implies -x_1 - x_2 + s_2 = 5$$

$$x_1, x_2, s_1 \geq 0$$

3. The objective function should be of maximization form. The minimization of a function $f(x)$ is equivalent to the maximization of the negative expression of the function $f(x)$.

Example

Min $Z = 3x_1 + x_2$

Ans: Max $Z' = -3x_1 - x_2$, where $Z' = -Z$

Question: Solve the following LPP using the simplex method.

Maximize $z = 12x_1 + 16x_2$

subjected to $10x_1 + 20x_2 \leq 120$

$8x_1 + 8x_2 \leq 80$

x_1 & $x_2 \geq 0$.

Solution:

Step 1: Convert inequalities into equalities by introducing slack variables.

$$10x_1 + 20x_2 + S_1 = 120$$

$$8x_1 + 8x_2 + S_2 = 80$$

$$x_1, x_2, S_1, S_2 \geq 0.$$

The objective function is rewritten as.

$$z = 12x_1 + 16x_2 + 0.S_1 + 0.S_2$$

Step 2: Intial simplex Table (Iteration 0).

- Let c_j denote coefficients of objective function; (12, 16, 0, 0), be first row.
- cb_i: denote coefficients of basic variables (slack / surplus), be first column.
- 2nd column be slack / surplus variables
- 2nd row be all basic variables.

Key column

cb_i	c_j	12	16	0	0		
	Basic Variable	x_1	x_2	s_1	s_2	RHS	Ratio
0	s_1	10	20	1	0	120	$\frac{120}{20} = 6$ min
0	s_2	8	8	0	1	80	$80/8 = 10$
$1 \le j \le 2$,	z_j	0	0	0	0		
	$c_j - z_j$	12	16	0	0		

Key row

- Remaining rows are coefficients of each variable given in step 1.

- Row z_j is defined as $z_j = \sum_{i=1}^{2} (cb_i)(a_{ij})$

- Last row is $c_j - z_j$.

- For maximization problem, optimal condition is all $c_j - z_j \leq 0$

- Since optimal condition is not reached. Select maximum value in $c_j - z_j$ and select that column. This column is called "key column".

- Take ratio of RHS and coefficient in the key column.

- Select the minimum value among ratios & select the row. This row is called "key row".

- The intersecting element is called "key element", which is 20 here.

- The variable x_2 is called entering variable & s_1 is called leaving variable

Step 3 : Construct table for Iteration - I.

- Since s_1 is leaving variable & x_2 is entering variable, replace s_1 by x_2 in the basic variable column

- Replace cb_1 accordingly (coefficient of new basic variable).

	C_j	12	16	0	0		
Cb.	B.V	x_1	x_2	s_1	s_2	RHS	Ratio
16	x_2	$\frac{1}{2}$	1	$\frac{1}{20}$	0	6	12
0	s_2	4	0	$-\frac{2}{5}$	1	32	8
	z_j	8	16	$\frac{4}{5}$	0		
	$C_j - Z_j$	4	0	$-\frac{4}{5}$	0		

$$\frac{1}{2} - \frac{\frac{1}{2} \times 4}{4} = 0$$

- To get new entries in row 1, divide each entry by "key element".

old

- For next row, consider the basic variable s_2 & its coefficient.

- The values in corresponding row is given by.

$$\text{new value} = \text{old value} - \frac{\text{Corr. key column value} \times \text{corr. key row value}}{\text{key element}}$$

$\therefore$ 1. $8 - \frac{8 \times 10}{20} = 8 - \frac{8\cancel{0}}{2\cancel{0}} = 8 - 4 = 2$

2. $8 - \frac{8 \times 20}{20} = 0$

3. $0 - \frac{1 \times 8}{20} = -\frac{2}{5}$

4.

- Now repeat previous step!

That is :

- Compute z_j and then $c_j - z_j$

$$z_j = \sum_{i=1}^{2} (cb_i)(a_{ij})$$

- For maximization problem, optimal condition is all $c_j - z_j \leq 0$.

- Since optimal condition is not reached. Select maximum value in $c_j - z_j$ and select that column. This column is called "key column".

- Take ratio of RHS and coefficient in the key column.

- Select the minimum value among ratios & select the row. This row is called "key row".

- The intersecting element is called "key element", which is 4 here.

- The variable x_1 is called entering variable & s_2 is called leaving variable.

Step 4 : Construct table for Iteration - II

	C_j	12	16	0	0	RHS
Cb_i	BV	x_1	x_2	s_1	s_2	
16	x_2	0	1	$1/10$	$-1/8$	2
12	x_1	1	0	$-1/10$	$1/4$	8
	Z_j	12	16	$2/5$	1	128
	$C_j - Z_j$	0	0	$-2/5$	-1	

Note that for maximization problem, optimality is reached if $C_j - Z_j \leq 0$.

$\therefore \quad x_1 = 8 \; ; \; x_2 = 2$

$\underline{\underline{Z = 128}}$

Minimization problem :

Minimize $z = x_1 - 3x_2 + 3x_3$ subject to

$$3x_1 - x_2 + 2x_3 \leq 7$$
$$2x_1 + 4x_2 \geq -12$$
$$-4x_1 + 3x_2 + 8x_3 \leq 10.$$

Solution: LPP in the standard form is

Maximize, $z^* = -x_1 + 3x_2 - 3x_3$

Multiply -1 to $2x_1 + 4x_2 \geq -12$, then

$$-2x_1 - 4x_2 \leq 12.$$

Step 1: Convert inequalities into equalities by introducing slack variables.

$$3x_1 - x_2 + 2x_3 + s_1 = 7$$
$$-2x_1 - 4x_2 + s_2 = 12$$
$$-4x_1 + 3x_2 + 8x_3 + s_3 = 10$$
$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0.$$

The objective function can be rewritten as $z^* = -x_1 + 3x_2 - 3x_3 + 0.S_1 + 0.S_2 + 0.S_3$

Step 2: Initial Simplex Table (Iteration 0).

	C_j	-1	3	-3	0	0	0			
C_{bi}	B.V	x_1	x_2	x_3	S_1	S_2	S_3	RHS	Ratio	
0	S_1	3	-1	2	1	0	0	7	—	
0	S_2	-2	-4	0	0	1	0	12	—	
0	S_3	-4	3	8	0	0	1	10	$10/3$	min
	Z_j	0	0	0	0	0	0	0		
	$C_j - Z_j$	-1	3	-3	0	0	0			

Entry variable is x_2 & leaving variable is S_3

	C_j	-1	3	-3	0	0	0		
C_{bi}	B.V	x_1	x_2	x_3	S_1	S_2	S_3	RHS	Ratio
0	S_1	$5/3$	0	$14/3$	1	0	$1/3$	$31/3$	$31/5$
0	S_2	$-22/3$	0	$32/3$	0	1	$4/3$	$76/3$	—
3	x_2	$-4/3$	1	$8/3$	0	0	$1/3$	$10/3$	—
	Z_j	-4	3	8	0	0	1	10	
	$C_j - Z_j$	3	0	-11	0	0	-1		

Entry variable is x_1 & leaving variable is S_1

	C_j	-1	3	-3	0	0	0	
C_{bi}	B.V	x_1	x_2	x_3	S_1	S_2	S_3	RHS
-1	x_1	1	0	$14/5$	$3/5$	0	$1/5$	$31/5$
0	S_2	0	0	$156/5$	$22/5$	1	$14/5$	$354/5$
3	x_2	0	1	$32/5$	$4/5$	0	$3/5$	$58/5$
	Z_j	-1	3	$82/5$	$9/5$	0	$8/5$	$143/5$
	$C_j - Z_j$	0	0	$-97/5$	$-9/5$	0	$-8/5$	

As $C_j - Z_j \le 0$, the solution is optimal

at $x_1 = \frac{31}{5}$; $x_2 = \frac{58}{5}$; $x_3 = 0$

With $\max z^* = \frac{143}{5}$ $\therefore$ $\min z = -\frac{143}{5}$.

Example

1. Using simplex method, Maximize

$$Z = 3x_1 + 5x_2 + 4x_3$$

subject to

$$2x_1 + 3x_2 \leq 8$$

$$2x_2 + 5x_3 \leq 10$$

$$3x_1 + 2x_2 + 4x_3 \leq 15$$

$$x_1, x_2 \geq 0$$

Step 1: Convert inequalities into equalities by introducing slack variables.

$$\therefore \quad 2x_1 + 3x_2 + s_1 = 8$$
$$2x_2 + 5x_3 + s_2 = 10$$
$$3x_1 + 2x_2 + 4x_3 + s_3 = 15$$

$$s_1, s_2, s_3 \geq 0$$

The objective function is rewritten as.

$$Z = 3x_1 + 5x_2 + 4x_3 + 0s_1 + 0s_2 + 0s_3.$$

Step 2: Intial simplex Table (Iteration 0).

- Let c_j denote coefficients of objective function; $(3, 5, 4, 0, 0, 0)$, be first row.
- cb_i: denote coefficients of basic variables (slack / surplus), be first column.

- 2^{nd} column be slack / surplus variables
- 2^{nd} row be all basic variables.

	c_j	3	5	4	0	0	0			min value
cb_i	Basic Variable	x_1	x_2	x_3	S_1	S_2	S_3	RHS	Ratio	↑
0	S_1	2	3	0	1	0	0	8	$8/3$	= 2.
0	S_2	0	2	5	0	1	0	10	$10/2$	= 5.
0	S_3	3	2	4	0	0	1	15	$15/2$	= 7..
$1 \le j \le 3$,	z_j	0	0	0	0	0	0	0		
	$c_j - z_j$	3	5	4	0	0	0	0		
			↓ max value							

- Remaining rows are coefficients of each variable given in step 1.
- Row z_j is defined as $z_j = \sum_{i=1}^{3} (cb_i)(a_{ij})$
- Last row is $c_j - z_j$
- For maximization problem, optimal condition is all $c_j - z_j \le 0$

- For minimization problem, optical condition is all $c_j - z_j \geq 0$.

- Since optimal condition is not reached. Select maximum value in $c_j - z_j$ and select that column. This column is called "key column".

- Take ratio of RHS and coefficient in the key column.

- Select the minimum value among ratios & select the row. This row is called "key row".

- The intersecting element is called "key element", which is 3 here.

- The variable x_2 is called entering variable & s_1 is called leaving variable

Step 3 : Construct table for Iteration - I.

- Since s_1 is leaving variable & x_2 is entering variable, replace s_1 by x_2 in the basic variable column

- Replace cb_1 accordingly (coefficient of new basic variable).

	c_j	3	5	4	0	0	0		
cb_i	Basic variables	x_1	x_2	x_3	s_1	s_2	s_3	RHS	Ratio
5	x_2	$2/3$	$3/3$	$0/3$	$1/3$	$0/3$	$0/3$	$8/3$	–
0	s_2	$-4/3$	0	5	$-2/3$	1	0	$14/3$	$14/15 \rightarrow$ min
0	s_3	$5/3$	0	4	$-2/3$	0	1	$29/3$	$29/12$
	z_j	$10/3$	$15/3$	0	$5/3$	0	0	$40/3$	–
	$c_j - z_j$	$-1/10$	0	4	$-5/3$	0	0		
				$\downarrow$ max					

- To get new entries in row 1, divide each entry by "key element".

- For next row, consider the basic variable S_2 & its coefficient.

- The values in corresponding row is given by.

$$\text{new value} = \text{old value} - \frac{\text{corr. key column value} \times \text{corr. key row value}}{\text{key element}}$$

$\therefore$ Entries are:

1. $0 - \dfrac{2 \times 2}{3} = -\dfrac{4}{3}$.

2. $2 - \dfrac{2 \times 3}{3} = 0$

3. $5 - \dfrac{0 \times 2}{3} = 5$

4. $0 - \dfrac{1 \times 2}{3} = -2/3$

- <u>Now repeat previous step!</u>

That is:

- Compute z_j and then $c_j - z_j$

$$z_j = \sum_{i=1}^{3} (cb_i)(a_{ij})$$

$$z_1 = (cb_1)(a_{11}) + (cb_2)(a_{21}) + (cb_3)(a_{31})$$

$$= 5 \times 2/3 + 0 + 0 = 10/3$$

- For maximization problem, optimal condition is all $c_j - z_j \leq 0$.

- Since optimal condition is not reached. Select maximum value in $c_j - z_j$ and select that column. This column is called "key column".

- Take ratio of RHS and coefficient in the key column.

- Select the minimum value among ratios & select the row. This row is called "key row".

- The intersecting element is called "key element", which is 5 here.

- The variable x_3 is called entering variable & s_2 is called leaving variable

<u>Step 4</u> : Construct table for Iteration - II

	C_j	3	5	4	0	0	0		
C_{bi}	Basic Variable	x_1	x_2	x_3	S_1	S_2	S_3	RHS	Ratio
5	x_2	$2/3$	1	0	$1/3$	0	0	$8/3$	4
4	x_3	$-4/15$	0	$5/5$	$-2/15$	$1/5$	0	$14/15$	$-4/14$ → skip
0	S_3	$41/15$	0	0	$-2/15$	$-4/3$	1	$89/15$	$\frac{89}{41}$
	Z_j	$34/15$	5	4	$17/15$	$4/5$	0	$\frac{256}{15}$	
	$C_j - Z_j$	$11/15$	0	0	$-17/15$	$-4/5$	0		

Leaving variable is s_3 & entry variable is x_1

Step 4: Construction table for iteration III.

	c_j	3	5	4	0	0	0	
cb_i	BV	x_1	x_2	x_3	s_1	s_2	s_3	RHS
5	x_2	0	1	0	$15/41$	$8/41$	$-10/41$	$50/41$
4	x_3	0	0	1	$-6/41$	$5/41$	$4/41$	$62/41$
3	x_1	1	0	0	$-2/41$	$-12/41$	$15/41$	$89/41$
	z_j	3	5	4	$45/41$	$24/41$	$11/41$	$765/41$
	$c_j - z_j$	0	0	0	$-45/41$	$-24/41$	$-11/41$	

Note that $c_j - z_j \leq 0$ $\therefore$ Optimal points is at $x_1 = \frac{89}{41}$, $x_2 = \frac{50}{41}$; $x_3 = \frac{62}{41}$ and

maximum value is $\frac{765}{41}$.